

KHALIL BOU KHALIL

Junior AI Engineer | LLM Applications, AI Agents, RAG and Machine Learning

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khalil-boukhalil.github.io/Portfolio

PROFILE

AI Master's candidate at CNAM Paris building end-to-end AI applications: LLM agents, RAG pipelines, FastAPI backends and machine learning models. Currently developing CTCARCH's internal multi-model AI platform. **Seeking a full-time AI Engineer position in France, available from October 2026.**

EDUCATION

- CNAM Paris – **Master 1 & Master 2, Artificial Intelligence for Connected Industries (AI4CI)** · Paris, France 2024 – 2026
- Lebanese International University (LIU) – **BSc in Computer Science** · Koura, Lebanon 2021 – 2024

PROFESSIONAL EXPERIENCE

- CTCARCH Pty Ltd – **AI Engineer Intern** · Remote, company based in Sydney, Australia Jul 2026 – Present
- Analyse workflows across architecture, construction, estimating and operations to identify practical AI use cases.
 - Architect and develop the foundation of an internal multi-model AI platform that centralises company knowledge and delivers department-specific assistants to around 12 employees (Python, FastAPI, Next.js, PostgreSQL, Docker).
 - Build the RAG and document-processing pipeline: PDF, Word and Excel ingestion, parsing, chunking, embeddings and pgvector search, powering document question answering with project context.
 - Implement the orchestration flow – task classification, retrieval, prompt construction, automatic model routing across OpenAI, Anthropic and Google APIs, human review – with routing driven by task type, quality, speed, cost and privacy.
 - Specify the governance layer under development: role-based access control, permissions, employee quotas, token and cost tracking, audit logs and an administrator dashboard.
- Safo Systems – **AI Agents Developer Intern** · Tripoli, Lebanon Jul 2025 – Sep 2025
- Developed Python multi-agent workflows automating document generation, with drafting, validation and human approval.
 - Applied LLM APIs, prompt engineering, tool calling and backend integration, and refined agent behaviour for reliability.
- MA Tech – **Full-Stack Developer Intern** · Minieh-Dannieh, Lebanon Jun 2024 – Sep 2024
- Contributed to full-stack React.js and Node.js applications: frontend interfaces, dashboards, backend services and APIs.
 - Participated in testing, debugging, performance improvements and Git-based development workflows.

SELECTED TECHNICAL PROJECTS

- DocuRAG – **Document Question-Answering Platform** | Python · FastAPI · React · FAISS · SQLAlchemy · Docker
- Built a full-stack RAG application combining FAISS vector search and BM25 into a hybrid retrieval pipeline over sentence-transformer embeddings, returning answers with source citations.
 - Added JWT authentication, SQLAlchemy persistence, rate limiting and Docker packaging over Ollama or LLM APIs.
- GenAI Multi-Agent Automation System | Python · LLM APIs · FastAPI · Streamlit
- Designed a three-agent workflow for data extraction, memo generation and human-in-the-loop validation in Python.
 - Exposed the workflow through a FastAPI and Streamlit interface for non-technical review and approval.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, Bash

LLM Engineering: LLM APIs (OpenAI, Anthropic, Google), AI agents, RAG, prompt engineering, embeddings, vector search

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, model training and evaluation, data preprocessing

Backend & Data: FastAPI, Flask, Node.js, REST APIs, authentication, RBAC, PostgreSQL, pgvector, FAISS, MongoDB

Frontend & DevOps: React, Next.js, Docker, Git, GitHub, Linux

LANGUAGES

Arabic – Native | English – B2, professional working proficiency | French – B1, intermediate